STAIR
DIGITAL

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# STAIR x Scaler School of Technology

**Date: 22nd April 2026**

## Instructions to Candidates

You are required to select **one** of the following problem statements and build a **working agent system**.

Your submission must include:

- A **short technical note** (architecture, decisions, trade-offs)
- **Test instructions** for evaluators
- Demo video
- A **deployed interface** (Optional)

Your system will be evaluated on:

- Correctness and reliability
- Depth of agent design
- Quality of outputs
- Observability and testability
- Practical deployment thinking

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## Task 1: Narrative-to-Visual Story Agent (10 Marks)

## Objective

Build an agent that converts a narrated story into a **structured visual output** (script + scene plan + generated video or animation).

## Requirements

- Accept **free-form narrative input** (text or voice converted to text)
- Convert into:
  - Structured script (scenes, dialogues, transitions)
  - Visual plan (scene descriptions, style, pacing)
- Generate a **visual output**:

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- Video OR
- Sequential storyboard frames (acceptable if video generation is constrained)

## Evaluation Criteria

- Coherence between input story and generated output
- Quality of scene breakdown and storytelling logic
- Ability to handle ambiguous or unstructured input
- Determinism / reproducibility of outputs

## Testability Requirement

You must provide:

- At least **3 predefined input stories**
- Expected characteristics of output (not exact output)
- Clear instructions to reproduce results
- Additionally, we may choose to provide a story and require seeing the video output, live which will be assessed on the aforementioned evaluation criteria.

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## Task 2: Source-Constrained Newsletter Generation Agent (5 Marks)

## Objective

Build an agent that generates a **highly referenced newsletter** strictly from a **defined set of sources**.

## Requirements

- Input: Topic (e.g., "AI regulation in India")
- Restrict knowledge to:
  - A predefined corpus (PDFs, reports, articles, etc.)
- Output must include:
  - Structured newsletter (sections, arguments)
  - Explicit citations:
    - Source name
    - Date of publication
    - Date of retrieval
    - Reference location (page / section)

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## Evaluation Criteria

- Strict adherence to source boundaries (no hallucinated knowledge)
- Depth and synthesis across multiple sources
- Quality and traceability of citations
- Handling of conflicting information

## Testability Requirement

You must:

- Provide the **exact dataset used**
- Demonstrate that removing a source changes the output
- Include at least **2 test prompts** and expected behavior
- We will provide examples (in PDF format) to test the effectiveness of the bot on the evaluation criteria.

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# Task 3: PDF-Constrained Conversational Agent (10 Marks)

## Objective

Build an agent that can **chat with a user about a given PDF**, with strict grounding.

## Requirements

- Accept any PDF as input
- Enable conversational querying
- The agent must:
  - Answer **only from the PDF**
  - Explicitly refuse out-of-scope queries
- Responses must include:
  - Citations (page number/section reference)

## Evaluation Criteria

- Accuracy of responses relative to the source
- Robustness against hallucination
- Quality of refusal when information is unavailable
- Retrieval and grounding quality

## Testability Requirement

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You must:

- Provide a **sample PDF**
- Include:
  - 5 valid queries
  - 3 invalid / out-of-scope queries
- Show expected behavior for both
- We will test out its effectiveness with a sample PDF to test the bot on the aforementioned evaluation criteria.

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## Bonus (Applicable to All Tasks)

Additional credit will be given if:

- The system supports **multiple languages**
- Demonstrates consistent performance across languages
- Maintains grounding and structure in non-English outputs

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## Alternate Track (Non-Technical / Business-Oriented Candidates)

## Task 4: Strategic Assessment of a Private Defence Company (5 Marks)

### Objective

Develop a **deep analytical assessment** of a private defence company, supported by structured reasoning.

### Requirements

- Deliver a **presentation (deck format)** covering:
  - Company overview and capabilities
  - Position within defence ecosystem
  - Competitive landscape
  - Revenue drivers and cost structures
  - Pricing dynamics and margin analysis
  - Forward-looking projections
- Incorporate:
  - Data-backed insights
  - Assumption transparency
  - Scenario-based thinking

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**Evaluation Criteria**

- Depth of analysis (not surface-level research)
- Clarity of strategic thinking
- Ability to connect technical capabilities with business outcomes
- Quality of structured communication

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**Final Note to Candidates**

This assessment is designed to evaluate your ability to:

- Translate ambiguity into structured systems
- Build reliable, testable AI workflows
- Operate at both **technical and problem-solving depth**

Superficial implementations will not score well, even if visually appealing.

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